

AVVENIRE
TECHNOLOGIES

INFINITRA PATENT FILE

Prepared by: Avvenire Technologies Inc.
Headquarters: Toronto, Ontario, Canada

FIG. 1 – Multi-Domain Operating System (100)

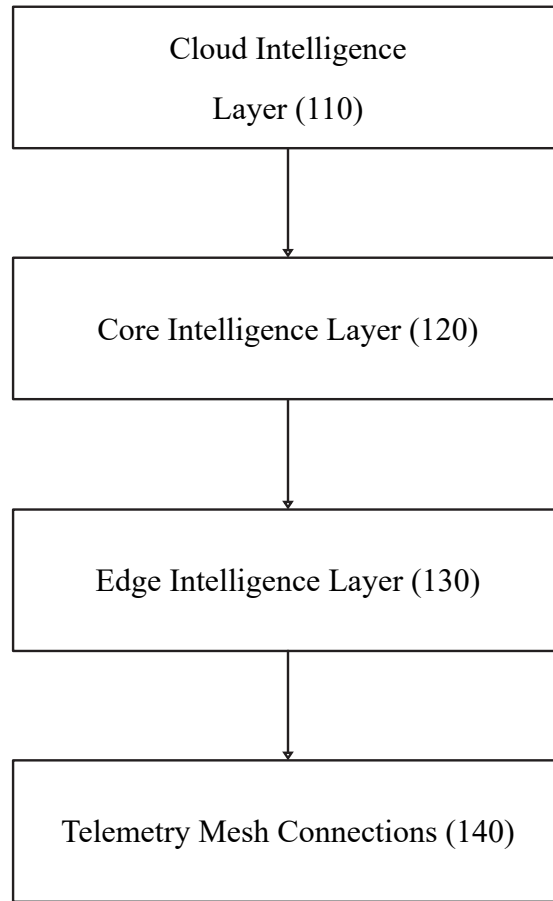


FIG. 2 – Tri-Layer Intelligence Model (200)

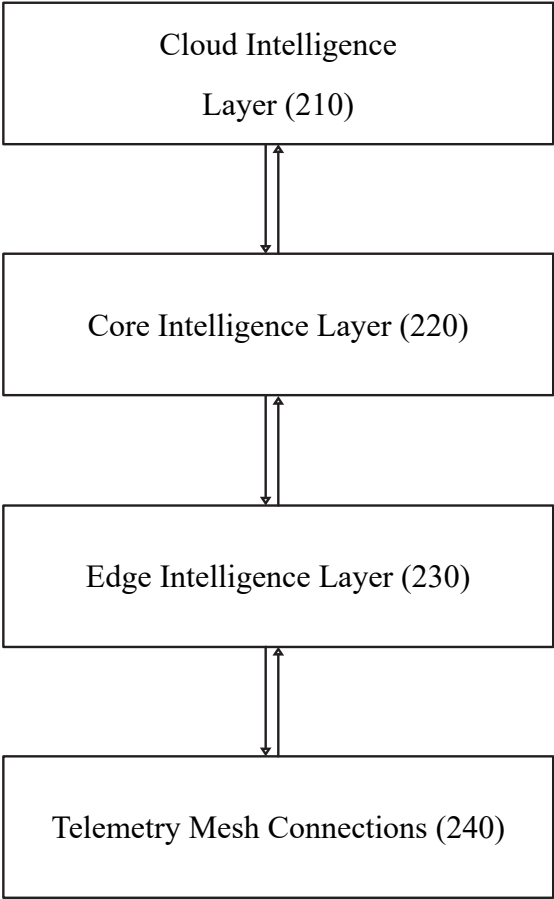


FIG. 3 – Distributed Telemetry Mesh (300)

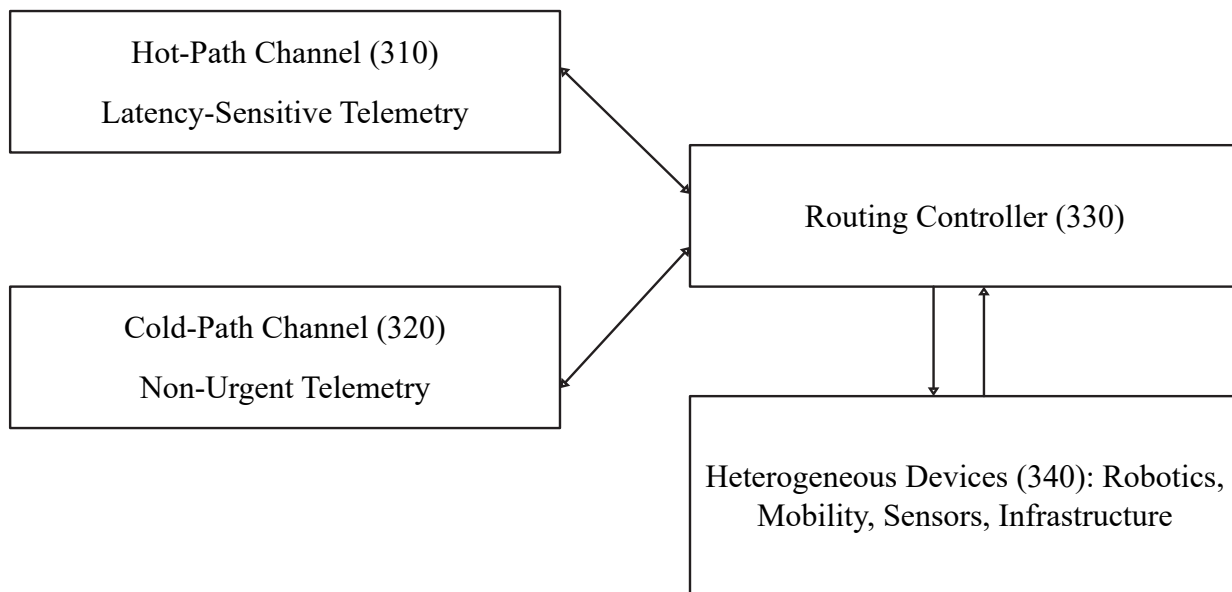


FIG. 4 – Multi-Agent Orchestration Subsystem (400)

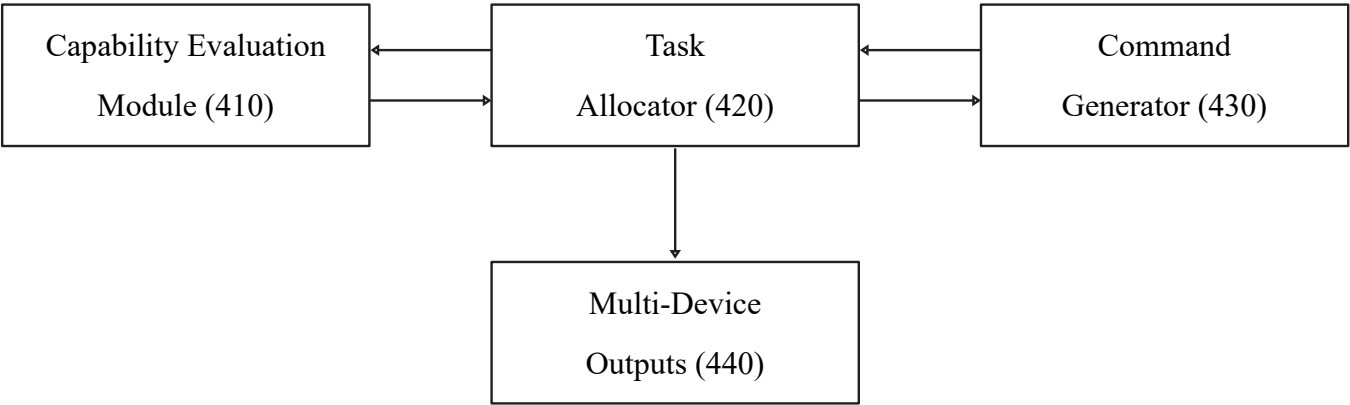


FIG. 5 – Arbitration Engine (500)

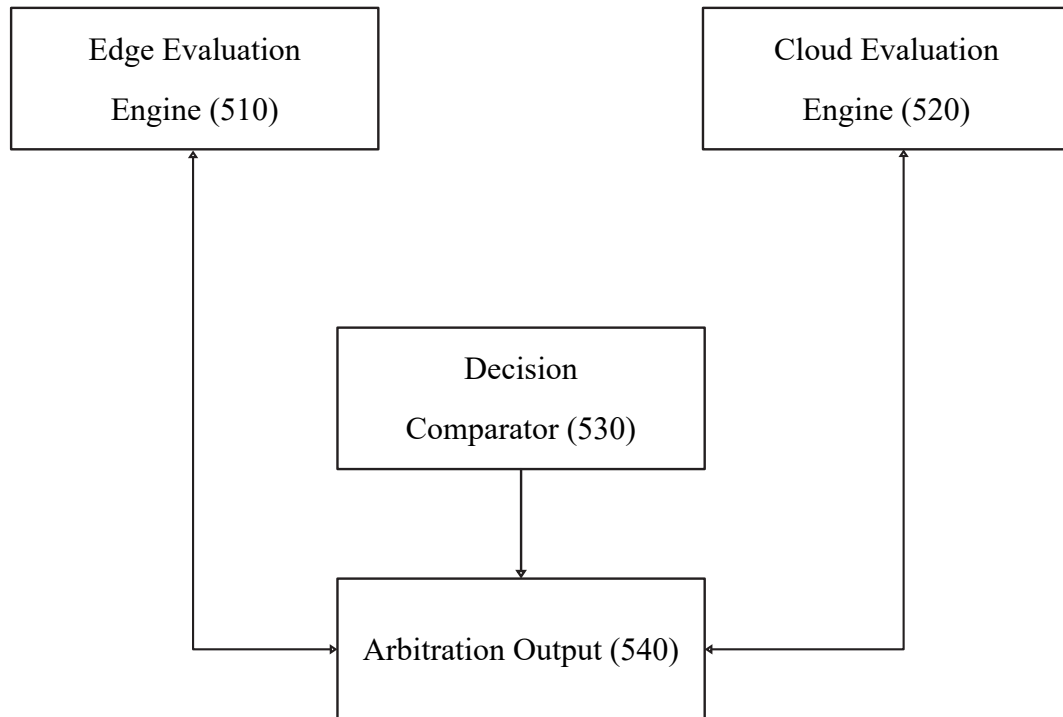


FIG. 6 – Safety-State Governance Subsystem (600)

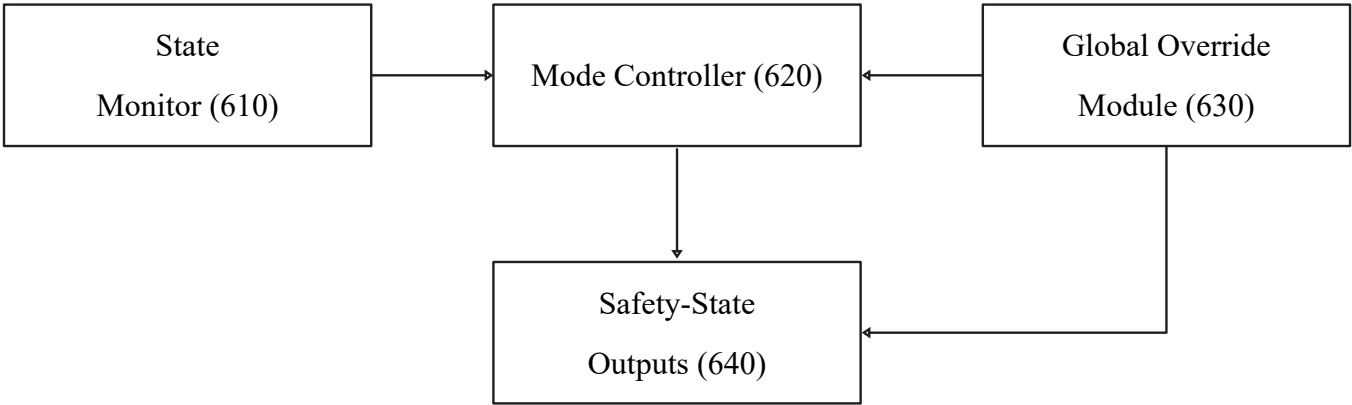
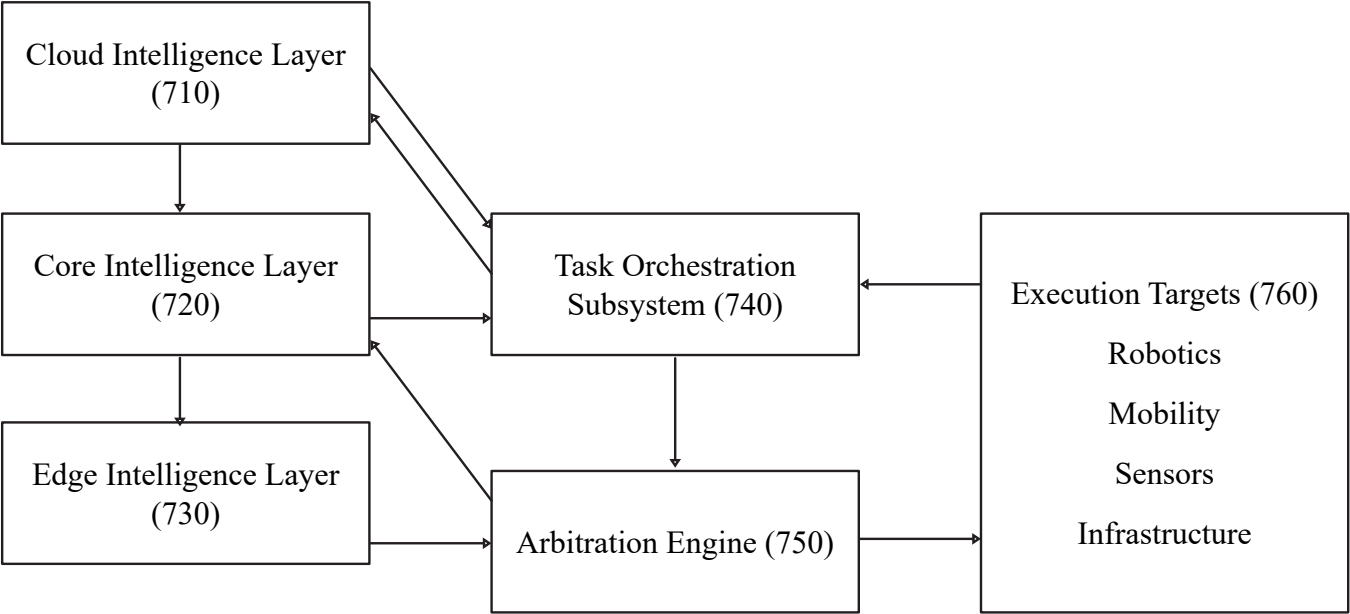


FIG. 7 – Multi-Domain Execution and Routing Flow (700)



ABSTRACT

A multi-domain intelligence operating system is disclosed, comprising cloud, core, and edge computational layers configured to perform long-horizon, mid-horizon, and real-time reasoning. A telemetry mesh transports device-generated data through differentiated hot-path and cold-path channels, enabling prioritized routing under dynamic conditions. An orchestration subsystem evaluates heterogeneous device capabilities, allocates tasks, and generates coordinated command outputs. An arbitration engine compares cloud and edge inference results to determine an optimal execution environment based on latency, accuracy, or resource availability. A safety-state governance subsystem detects hazards, transitions the system between operating modes, and enforces overrides during emergencies. A fusion subsystem integrates internal telemetry with external contextual information to form a unified knowledge representation consumable by all intelligence layers. Together, these components provide predictive, resilient, and synchronized operation across distributed device fleets.

Claims

1.

A multi-domain operating system comprising:

- (a) a cloud intelligence layer configured to execute long-horizon reasoning operations;
 - (b) a core intelligence layer configured to execute mid-horizon reasoning operations;
 - (c) an edge intelligence layer configured to execute real-time reasoning operations on heterogeneous devices;
 - (d) a telemetry mesh comprising a hot-path channel for latency-sensitive data and a cold-path channel for non-urgent data;
 - (e) an orchestration subsystem comprising a capability evaluation module, a task allocator, and a command generator configured to assign tasks to heterogeneous devices;
 - (f) an arbitration engine comprising an edge evaluation engine, a cloud evaluation engine, and a decision comparator configured to select an execution environment for a computational workload; and
 - (g) a safety-state governance subsystem comprising a state monitor, a mode controller, and an override module configured to enforce safety constraints during hazard conditions;
- wherein the cloud, core, and edge intelligence layers operate collaboratively to provide synchronized reasoning across multiple temporal horizons.

2.

The system of claim 1, wherein the telemetry mesh dynamically reallocates bandwidth between the hot-path and cold-path channels based on environmental risk or predicted hazard levels.

3.

The system of claim 1, wherein a routing controller prevents priority inversion by ensuring cold-path data does not obstruct hazard-related telemetry.

4.

The system of claim 1, wherein the orchestration subsystem generates multi-device command sequences including timing markers, fallback rules, or rollback conditions.

5.

The system of claim 1, wherein the arbitration engine selects the execution environment based on latency, accuracy, model confidence, environmental safety, or network conditions.

6.

The system of claim 1, wherein the safety-state governance subsystem transitions the

system between normal, degraded, and emergency modes using multi-level hazard thresholds.

7.

The system of claim 1, wherein the override module supersedes orchestration and arbitration outputs during emergency mode.

8.

The system of claim 1, further comprising a fusion subsystem configured to generate a unified knowledge representation from internal telemetry inputs and external contextual inputs.

9.

The system of claim 8, wherein unified knowledge representations are consumable by any intelligence layer for cross-layer reasoning.

10.

The system of claim 1, wherein the arbitration engine detects inference divergence or model drift by comparing cloud and edge evaluation outputs.

Method Claims

11.

A method for multi-domain reasoning comprising:

- (a) receiving telemetry from heterogeneous devices;
- (b) routing latency-sensitive telemetry through a hot-path channel and non-urgent telemetry through a cold-path channel;
- (c) performing long-horizon, mid-horizon, and real-time reasoning across cloud, core, and edge intelligence layers;
- (d) evaluating device capabilities;
- (e) allocating tasks and generating multi-device command outputs;
- (f) performing comparative arbitration between cloud and edge inference outputs; and
- (g) enforcing safety-state transitions using a safety-state governance subsystem.

12.

The method of claim 11, further comprising dynamically adjusting telemetry routing based on predicted hazard conditions.

13.

The method of claim 11, wherein arbitration includes computing a confidence gap between cloud and edge inference outputs.

14.

The method of claim 11, wherein safety-state enforcement operates independently of cloud connectivity.

Computer-Readable Medium

15.

A non-transitory computer-readable medium storing instructions which, when executed by one or more processors, cause the processors to perform the method of any of claims 11–14.

Additional System and Multi-Agent Claims

16.

The system of claim 1, wherein the telemetry mesh includes redundant routing paths for fault tolerance.

17.

The system of claim 1, wherein the capability evaluation module produces readiness scores based on sensor reliability, actuator condition, environmental context, or energy reserves.

18.

The system of claim 1, wherein the fusion subsystem performs temporal and spatial fusion of telemetry.

19.

The system of claim 1, wherein the arbitration engine proactively switches execution environments based on predicted network degradation.

20.

The system of claim 1, wherein multi-agent coordination includes hazard sharing across devices.

21.

The system of claim 1, wherein cooperative task execution includes synchronized movement, formation control, or distributed sensing.

22.

The system of claim 1, further comprising an external information interface supplying environmental models, hazard alerts, or operational directives.

23.

The system of claim 1, wherein fallback behaviors are triggered when connectivity is degraded.

Multi-Agent System Claims

24.

A distributed multi-agent coordination system comprising a plurality of heterogeneous devices configured to:

- (a) exchange hazard telemetry via a hot-path channel;
- (b) receive coordinated task allocations from an orchestration subsystem;
- (c) execute distributed command outputs; and
- (d) adjust formation or behavior based on safety-state transitions.

25.

The system of claim 24, wherein arbitration decisions incorporate aggregated telemetry from multiple devices.

26.

The system of claim 24, wherein coordinated search patterns include sweeps, spirals, boundary scans, or patrol formations.

27.

The system of claim 1, wherein long-horizon, mid-horizon, and real-time reasoning operate synergistically to provide emergent behaviors not achievable by single-layer architectures.

Infinitra Multi-Domain Intelligence Operating System with Telemetry Mesh, Orchestration, Arbitration, and Safety-State Governance

This invention concerns a multi-domain operating system enabling synchronized reasoning across cloud, core, and edge environments using a telemetry mesh, task-orchestration logic, dual-path arbitration mechanisms, and safety-state governance.

Technical Field

The present disclosure relates to distributed computing architectures and, in particular, to a multi-domain intelligence operating system configured to coordinate reasoning, telemetry transport, task orchestration, execution environment arbitration, safety-state governance, and multi-layer decision flow across cloud, core, and edge computational environments.

The disclosed system integrates telemetry-driven computation, real-time device coordination, and multi-horizon reasoning to improve the performance, reliability, and safety of heterogeneous device networks.

The invention further pertains to intelligent autonomous platforms, robotic fleets, sensor networks, mobility systems, and other distributed hardware environments requiring synchronized decision-making and low-latency operational control.

Background Art

Modern distributed computing systems frequently rely on single-domain architectures—such as cloud-only or edge-only computation—which limit scalability, increase latency during critical events, and reduce operational safety.

Conventional hybrid systems that combine cloud and edge computation typically do not provide coordinated reasoning across multiple temporal horizons and lack mechanisms for dynamically selecting an execution environment when operating conditions change.

Existing platforms also fail to implement differentiated telemetry pathways to segregate latency-sensitive data from non-urgent information, resulting in congestion, delayed hazard response, or inefficient bandwidth usage.

Known distributed architectures do not incorporate arbitration engines capable of comparing inference outputs produced at different computational layers to determine the most appropriate execution environment for a given task.

Existing safety mechanisms in distributed systems generally cannot supersede orchestration or arbitration logic during hazardous or degraded situations, leading to unpredictable behavior or unacceptable delays in safety-critical contexts.

Many systems lack the ability to integrate internal device telemetry with external contextual information into a unified knowledge representation suitable for consistent reasoning across distributed intelligence layers.

These technical limitations reduce robustness, increase latency during hazard events, prevent scalable multi-agent coordination, and limit the effectiveness of autonomous or semi-autonomous device networks.

Accordingly, there exists a need for a multi-domain intelligence operating system that supports synchronized reasoning, prioritized telemetry routing, distributed arbitration, coordinated orchestration, and deterministic safety-state transitions.

Disclosure of Invention — Technical Problem

Modern distributed computing architectures often rely on either cloud-only or edge-only infrastructures, resulting in performance bottlenecks, poor scalability, and inadequate responsiveness during time-critical events.

Existing hybrid cloud-edge systems do not provide a unified mechanism for determining where computation should occur when operating conditions fluctuate or when network degradation affects latency and decision integrity.

Known architectures fail to separate latency-sensitive telemetry from non-urgent data, causing priority inversion where critical hazard-related information is delayed by bulk or low-priority transmissions.

Conventional systems lack multi-horizon reasoning that synchronizes long-term planning, mid-horizon adaptation, and real-time actuation across distributed intelligence layers.

Existing platforms do not incorporate a coordinated orchestration subsystem capable of evaluating device readiness, assigning tasks, and generating multi-device command flows based on capability scoring.

No unified arbitration framework exists to compare inference outputs from different computational domains (cloud vs. edge) and select the environment best suited to the operational context.

Safety mechanisms in distributed systems rarely include a dedicated safety-state governance subsystem that can supersede orchestration or arbitration logic during hazardous events.

Current distributed intelligence architectures lack a mechanism to unify internal telemetry and external contextual information into a single knowledge representation consumable across cloud, core, and edge domains.

Disclosure of Invention — Solution to Problem

The disclosed invention addresses these issues by providing a multi-domain operating system comprising coordinated cloud, core, and edge intelligence layers that perform long-horizon, mid-horizon, and real-time reasoning, respectively.

A differentiated telemetry mesh transports data using a hot-path channel for latency-critical information and a cold-path channel for non-urgent telemetry, preventing priority inversion and ensuring timely hazard response.

The orchestration subsystem evaluates heterogeneous device capabilities, computes readiness scores, assigns appropriate tasks, and produces synchronized multi-device command sequences.

The arbitration engine compares inference results generated by cloud and edge computational layers and selects the optimal execution environment based on latency, accuracy, safety constraints, or network quality.

A safety-state governance subsystem monitors telemetry for anomalies, transitions the system between normal, degraded, and emergency modes, and applies overrides to enforce deterministic safety behavior.

A fusion subsystem integrates internal telemetry with external contextual information to produce unified knowledge representations usable across all intelligence layers.

The combined architecture enables scalable multi-agent coordination, predictive hazard response, robust operation under variable network conditions, and consistent decision-making across distributed environments.

Advantageous Effects of the Invention

The invention provides improved scalability, robustness, and responsiveness by enabling synchronized multi-horizon reasoning across cloud, core, and edge intelligence layers.

The telemetry mesh reduces latency during hazard events by segregating high-priority data into a dedicated hot-path channel, increasing system safety and improving actuator response times.

The arbitration engine reduces inference failures by comparing cloud-level and edge-level outputs, detecting divergence, and adaptively selecting the execution environment best suited to real-time operational constraints.

The orchestration subsystem enables efficient multi-agent coordination, allowing heterogeneous devices to cooperate on distributed tasks, support fallback operations, and maintain mission continuity during partial network degradation.

The safety-state governance subsystem ensures deterministic behavior during hazards by enforcing emergency overrides that supersede orchestration or arbitration logic, even when connectivity is impaired.

The combined architecture improves environmental awareness, prediction accuracy, device lifespan, and system-level reliability compared to single-domain or non-hierarchical distributed intelligence platforms.

Brief Description of Drawings

FIG. 1 illustrates a multi-domain operating system architecture comprising cloud, core, and edge intelligence layers interconnected through a telemetry mesh.

FIG. 2 illustrates a tri-layer intelligence model showing the distribution of long-horizon, mid-horizon, and real-time reasoning functions across the cloud, core, and edge layers.

FIG. 3 illustrates a distributed telemetry mesh comprising a hot-path channel for latency-sensitive data, a cold-path channel for non-urgent data, and a routing controller for dynamic allocation.

FIG. 4 illustrates an orchestration subsystem including a capability evaluation module, a task allocator, and a command generator configured for multi-device coordination.

FIG. 5 illustrates an arbitration engine comprising an edge evaluation engine, a cloud evaluation engine, and a decision comparator that selects execution environments.

FIG. 6 illustrates a safety-state governance subsystem comprising a state monitor, mode controller, and override module for emergency operations.

FIG. 7 illustrates a multi-domain execution and routing flow demonstrating how orchestration, arbitration, intelligence layers, and execution targets interact dynamically during operation.

Mode(s) for Carrying Out the Invention

The following detailed description is provided to enable a thorough understanding of the disclosed multi-domain operating system. The examples described herein illustrate representative embodiments and are not intended to limit the scope of the invention.

FIG. 1 depicts a multi-domain operating system comprising cloud, core, and edge intelligence layers interconnected through a telemetry mesh. These layers collaborate through synchronized reasoning, arbitration, and orchestration processes.

FIG. 2 illustrates a tri-layer intelligence model in which long-horizon reasoning operations are performed by the cloud intelligence layer, mid-horizon reasoning operations are carried out by the core intelligence layer, and real-time reasoning operations are executed by the edge intelligence layer.

Long-horizon reasoning includes strategic forecasting, mission planning, model refinement, and large-scale inference operations that typically require high compute capacity.

Mid-horizon reasoning includes policy propagation, contextual adjustments, regional aggregation, and coordination of device clusters.

Real-time reasoning includes local inference, actuator control, sensor fusion, and immediate decision-making executed on heterogeneous devices.

FIG. 3 illustrates a distributed telemetry mesh comprising a hot-path channel for latency-sensitive telemetry and a cold-path channel for non-urgent telemetry. A routing controller dynamically allocates bandwidth and directs data flow based on operational context.

The hot-path channel transports telemetry such as hazard-state data, actuator feedback, or high-priority environmental readings. This channel is optimized for minimal latency.

The cold-path channel transports telemetry such as historical logs, non-urgent metadata, or long-range environmental information.

FIG. 4 illustrates an orchestration subsystem comprising a capability evaluation module, task allocator, and command generator. This subsystem assigns tasks to devices and produces coordinated multi-device command outputs.

The capability evaluation module computes readiness scores for devices based on factors including energy reserves, processor load, thermal conditions, sensor confidence, and actuator status.

The task allocator distributes tasks to one or more devices, potentially forming coordinated multi-agent groups based on mission objectives.

The command generator produces execution sequences that may include timing constraints, synchronization markers, branching logic, or rollback conditions.

FIG. 5 illustrates an arbitration engine comprising an edge evaluation engine and a cloud evaluation engine. A decision comparator evaluates the outputs of these evaluation engines and selects an execution environment.

Selection criteria may include latency, accuracy, model confidence, environmental safety, or network quality conditions.

Arbitration may be triggered periodically, event-driven, or based on predictive thresholds derived from historical telemetry.

FIG. 6 illustrates a safety-state governance subsystem configured to manage system transitions into normal, degraded, and emergency operating modes.

The state monitor detects anomalies or hazards using multi-level thresholds.

The mode controller transitions the system between modes based on hazard severity, device condition, or environmental context.

In emergency mode, an override module supersedes orchestration and arbitration outputs to enforce safety constraints.

FIG. 7 illustrates a multi-domain execution and routing flow comprising cloud intelligence layer, core intelligence layer, and edge intelligence layer. A task orchestration subsystem coordinates workload distribution, while an arbitration engine evaluates execution contexts and determines whether tasks are performed in cloud, core, or edge environments.

Execution targets may include robotic platforms, mobility systems, sensors, or infrastructure components receiving final execution commands.

Internal telemetry may include sensor readings, device states, actuator feedback, or localized environmental data.

External information may include weather models, hazard alerts, regional constraints, map data, or mission metadata.

The unified knowledge representation produced by the fusion subsystem provides a consistent decision substrate used for learning, arbitration, and multi-agent coordination.

The intelligence layers operate synergistically to provide robust distributed reasoning. The cloud intelligence layer performs computationally intensive tasks; the core intelligence layer performs regional aggregation and contextual adaptation; and the edge intelligence layer provides fast, localized decision-making.

This tri-layer configuration enables simultaneous execution of long-horizon planning, mid-horizon policy refinement, and real-time responsiveness, producing emergent capabilities not achievable by conventional systems relying on a single computational domain.

Industrial Applicability

The disclosed multi-domain operating system is applicable to any environment requiring coordinated reasoning, distributed decision-making, or telemetry-driven actuation across heterogeneous devices. The system may be deployed in robotics, autonomous mobility fleets, industrial automation platforms, environmental monitoring networks, infrastructure management systems, and multi-agent operational domains.

The architecture supports large-scale installations involving thousands or millions of devices, each contributing telemetry to cloud, core, or edge intelligence layers. This enables wide applicability in commercial, industrial, governmental, and research settings requiring scalable computation and real-time responsiveness.

The system's tri-layer intelligence model, telemetry mesh, orchestration subsystem, arbitration engine, and safety-state governance subsystem provide deterministic reasoning behaviors, making the invention suitable for mission-critical applications such as logistics, transportation, emergency response, hazardous-environment robotics, and security systems.

The invention may also be applied to distributed sensor networks, manufacturing ecosystems, smart-city infrastructures, agricultural automation, energy-grid management, and autonomous inspection or maintenance systems where synchronized multi-agent coordination is required.

Reference Signs List

110 – Cloud Intelligence Layer

120 – Core Intelligence Layer

130 – Edge Intelligence Layer

140 – Telemetry Mesh

210 – Cloud Intelligence Layer (Tri-Layer Model)

220 – Core Intelligence Layer (Tri-Layer Model)

230 – Edge Intelligence Layer (Tri-Layer Model)

240 – Telemetry Mesh (Tri-Layer Model)

310 – Hot-Path Channel

320 – Cold-Path Channel

330 – Routing Controller
340 – Heterogeneous Devices (Telemetry Sources)

410 – Capability Evaluation Module
420 – Task Allocator
430 – Command Generator
440 – Multi-Device Command Outputs

510 – Edge Evaluation Engine
520 – Cloud Evaluation Engine
530 – Decision Comparator
540 – Arbitration Output

610 – State Monitor
620 – Mode Controller
630 – Global Override Module
640 – Safety-State Outputs

710 – Cloud Intelligence Layer (Execution Flow)
720 – Core Intelligence Layer (Execution Flow)
730 – Edge Intelligence Layer (Execution Flow)
740 – Task Orchestration Subsystem
750 – Arbitration Engine
760 – Execution Targets (Robotics, Mobility, Sensors, Infrastructure)